

CLARA WONG-FANNJIANG

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EDUCATION	University of California, Berkeley <i>Ph.D. in Electrical Engineering & Computer Sciences</i> Advisors: Michael I. Jordan and Jennifer Listgarten	Aug 2018 – Aug 2023 Berkeley, CA
	Stanford University <i>B.S. in Computer Sciences, with Honors and Distinction</i>	Sep 2012 – June 2016 Stanford, CA
PROFESSIONAL EXPERIENCE	Genentech <i>Senior Machine Learning Scientist</i> Research on uncertainty quantification frameworks for machine learning in drug discovery.	Sep 2023 – present South San Francisco, CA
	Salesforce Research <i>Research Intern</i> Developed methods for designing thermostable enzymes using protein language models with Ali Madani and Nikhil Naik.	May 2022 – Oct 2022 Palo Alto, CA
	Monterey Bay Aquarium Research Institute <i>Research Assistant</i> Led fieldwork to investigate and model <i>in situ</i> behavior of scyphozoans in Monterey Bay with Kakani Katija.	Feb 2018 – Aug 2018 Moss Landing, CA
	Google <i>Google Brain Resident</i> Research on chaotic dynamics in recurrent neural networks with David Susillo.	June 2016 – June 2017 Mountain View, CA
PREPRINTS	3. “Conformal policy control”. Drew Prinster, Clara Fannjiang, Ji Won Park, Kyunghyun Cho, Anqi Liu, Suchi Saria, and Samuel Stanton. arXiv:2603.02196. 2026. [arXiv] 2. “E-evaluator: reliable agent verifiers with sequential hypothesis testing.” Shuvom Sadhuka, Drew Prinster, Clara Fannjiang, Gabriele Scalia, Aviv Regev, and Hanchen Wang. arXiv:2512.03109. 2025. [arXiv] 1. “Deep evolutionary fitness inference for variant nomination from directed evolution.” M. W. Shen, N. Diamant, C. Helmling, R. Newland, Z. Lu, C. Fannjiang, S. Kelow, N. Frey, S. Saremi, R. Kelly, R. Bonneau, G. Scalia, C. Cunningham, and T. Biancalani. biorXiv: 2025.07.22.666175. 2025. [bioRxiv]	
PERSPECTIVES & PRIMERS	3. “Machine learning for protein engineering.” Kadina E. Johnston, Clara Fannjiang, Bruce J. Wittmann, Brian L. Hie, Kevin K. Yang, and Zachary Wu. In <i>Challenges and Advances in Computational Chemistry and Physics</i> (Springer), 277-311. 2023. [DOI] 2. “Is novelty predictable?” Clara Fannjiang and Jennifer Listgarten. <i>Cold Spring Harbor Perspectives in Biology</i>, 16, a041469. 2024. [DOI]	

JOURNAL
PUBLICATIONS

1. “Generative models for protein structures and sequences.” Chloe Hsu, **Clara Fannjiang**, and Jennifer Listgarten. *Nature Biotechnology* 42, 196–199. 2024. [\[DOI\]](#)
- * denotes equal contribution. α - β denotes alphabetical ordering.
7. “Optimal trade-off control in machine learning-based library design, with application to adeno-associated virus (AAV) for gene therapy.” Danqing Zhu*, David H. Brookes*, Akosua Busia*, Ana Carneiro, **Clara Fannjiang**, Galina Popova, David Shin, Tomasz J. Nowakowski, Jennifer Listgarten, and David. V. Schaffer. *Science Advances*, 10, eadj3786. 2024. [\[DOI\]](#)
6. “Prediction-powered inference.” (α - β) Anastasios N. Angelopoulos, Stephen Bates, **Clara Fannjiang**, Michael I. Jordan, and Tijana Zrnic. *Science*, 382, 669–674. 2023. [\[DOI\]](#)
5. “Conformal prediction under feedback covariate shift for biomolecular design.” **Clara Fannjiang**, Stephen Bates, Anastasios Angelopoulos, Jennifer Listgarten, and Michael I. Jordan. *Proceedings of the National Academy of Sciences*, 119(43), e2204569119. 2022. [\[DOI\]](#)
4. “Learning protein fitness models from evolutionary and assay-labelled data.” Chloe Hsu, Hunter Nisonoff, **Clara Fannjiang**, and Jennifer Listgarten. *Nature Biotechnology*, 40, 1114–1122. 2022. [\[DOI\]](#)
3. “Mobile robotic platforms for the acoustic tracking of deep-sea demersal fishery resources.” I. Masmitja, J. Navarro, S. Gomariz, J. Aguzzi, B. Kieft, T. O’Reilly, K. Katija, P. J. Bouvet, **C. Fannjiang**, M. Vigo, P. Puig, A. Alcocer, G. Vallicrosa, N. Palomeras, M. Carreras, J. Del-Rio, and J. B. Company. *Science Robotics*, 5(48), eabc3701. 2020. [\[DOI\]](#)
2. “Augmenting biologging with supervised machine learning to study in situ behavior of the medusa *Chrysaora fuscescens*.” **Clara Fannjiang**, T. Aran Mooney, Seth Cones, David Mann, K. Alex Shorter, and Kakani Katija. *Journal of Experimental Biology*, 222, jeb207654. 2019. [\[DOI\]](#)
1. “Optimal arrays for compressed sensing in snapshot-mode radio interferometry.” **Clara Fannjiang**. *Astronomy & Astrophysics*, 559, A73. 2013. [\[DOI\]](#)

REFEREED
CONFERENCE
PROCEEDINGS

5. “Reliable algorithm selection for machine learning-guided design.” **Clara Fannjiang** and Ji Won Park. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. [\[proceedings\]](#)
4. “Variational refinement for importance sampling using the forward Kullback-Leibler divergence.” Ghassen Jerfel*, Serena Wang*, **Clara Fannjiang**, Katherine Heller, Yian Ma, Michael Jordan. In *Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2021. [\[proceedings\]](#)
3. “Autofocused oracles for model-based design.” **Clara Fannjiang** and Jennifer Listgarten. In *Advances in Neural Information Processing Systems* 33 (NeurIPS), 2020. [\[proceedings\]](#)

WORKSHOP
PROCEEDINGS &
POSTERS

2. "A view of estimation of distribution algorithms through the lens of expectation-maximization." David H. Brookes, Akosua Busia, **Clara Fannjiang**, Kevin Murphy, and Jennifer Listgarten. In *Proceedings of the 2020 Genetic and Evolutionary Computation Conference (GECCO)*, 2020. [\[DOI\]](#)
1. "Area-only method for underwater object tracking using autonomous vehicles." Ivan Masmitja, Spartacus Gomariz, Joaquin Del Rio, Brian Kieft, Tom O'Reilly, Jacobo Aguzzi, Pierre-Jean Bouvet, **Clara Fannjiang**, and Kakani Katija. In *OCEANS 2019 – Marseille*, 2019. [\[DOI\]](#)
3. "Designing active and thermostable enzymes with sequence-only predictive models." **C. Fannjiang**, M. Olivas, E. Greene, C. J. Markin, B. Wallace, B. Krause, M. M. Pinney, J. S. Fraser, P. M. Fordyce, A. Madani, and N. Naik. NeurIPS Workshop on Learning Meaningful Representations of Life, 2022. [\[poster\]](#)
2. "Hallucinations in neural machine translation." Katherine Lee, Orhan Firat, Ashish Agarwal, **Clara Fannjiang**, and David Sussillo. NeurIPS Workshop on Interpretability and Robustness in Audio, Speech, and Language, 2018. [\[PDF\]](#)
1. "Pushing the boundaries of fine-grained object classification with fMRI decoding in human occipito-temporal cortex." **Clara Fannjiang**, Coraline Rinn Iordan, Diane Beck, and Fei-Fei Li. Vision Science Society Annual Meeting (VSS), 2015. [\[abstract\]](#)

TALKS

16. "Reliable algorithm selection for machine learning-guided design." Invited talk, Berkeley-Genentech Workshop on Uncertainty Quantification for Decision-Making, October 2025.
15. "Reliable algorithm selection for machine learning-guided design." Contributed talk, Machine Learning in Computational Biology (MLCB) Meeting, 2025.
14. "Reliable algorithm selection for machine learning-guided design." Guest lecture, UC Berkeley COMPSCI 294-150 (AI Meets Biology & Chemistry), April 2025.
13. "Error embraced: Making trustworthy scientific decisions with imperfect predictions." Invited talk, Simons Institute Workshop on AI≡Science: Strengthening the Bond Between the Sciences and AI, June 2024.
12. "Prediction-powered inference." Guest lecture, UC Berkeley COMPSCI 294-150 (AI Meets Biology & Chemistry), February 2024.
11. "Prediction-powered inference." Invited talk, ETH AI Center Machine Learning for Health Care and Life Sciences Workshop, November 2023.
10. "Prediction-powered inference." Invited talk, Alaa Lab Rising Star Series, September 2023.
9. "Prediction-powered inference." Invited talk, Calico Life Sciences, May 2023.
8. "Prediction-powered inference." Invited talk, Microbiology Society Annual Conference, April 2023.
7. "Conformal prediction for biomolecular design." Invited talk, EPFL Physics of Living Systems Seminar Series, November 2022.

6. "Machine learning-guided tour of thermostability and activity in two enzyme families." Invited talk, James Fraser Lab, October 2022.
5. "Conformal prediction for the design problem." Invited talk, Machine Learning for Protein Engineering (ML4PE) Seminar Series, October 2022.
4. "Conformal prediction for the design problem." Invited talk, AIDD Summer School on Advanced Machine Learning for Drug Discovery, May 2022.
3. "Autofocused oracles for model-based design." Invited talk, Machine Learning to Accelerate Biology (MLAB) Series, August 2020.
2. "Using machine learning to deduce fine-scale behavior of jellyfish (*Chrysaora fuscescens*) in Monterey Bay." Contributed talk (**Marlene Zuk Award for Best Student Presentation Finalist**), Society for Integrative and Comparative Biology Annual Meeting, January 2019.
1. "Augmenting biologging with supervised machine learning to study *in situ* behavior of the medusa *Chrysaora fuscescens*." Invited talk, Monterey Bay Aquarium Research Institute Seminar, 2018.

TEACHING
EXPERIENCE

University of California, Berkeley

STAT 102: Data, Inference, & Decisions. Graduate Student Instructor & Guest Lecturer, Fall 2020.

STAT 102: Data, Inference, & Decisions. Graduate Student Instructor. Spring 2020.

Stanford University

EE 364A: Convex Optimization. Teaching Assistant, Winter 2016.

EE 103: Introduction to Matrix Methods. Teaching Assistant, Fall 2015.

SERVICE

Organizing & Outreach

Organizer, Berkeley-Genentech Workshop on Uncertainty Quantification for Decision-Making, 2025

Coordinator, UC Berkeley Women in Computer Science and Electrical Engineering (WiCSE), 2019-2020

Mentor, Berkeley AI Research Undergraduate Mentoring Program, 2019-2020

Instructor, Stanford Splash! B4329: The Biology of Vision and Perception, 2015

Instructor, Stanford Splash! M4053: For the Love of Optimization, 2014

Organizer, Stanford SAILORS (now Stanford AI4ALL), 2014

Reviewing

Area Chair, Machine Learning for Health (ML4H) Symposium, 2025

Reviewer for *Journal of the Royal Statistical Society: Series C*, International Conference on Machine Learning (ICML), and Conference on Neural Information Processing Systems (NeurIPS).

AWARDS

NSF Graduate Research Fellowship, 2019.

SICB Marlene Zuk Best Student Presentation Finalist, 2019.

UC Berkeley Allen D. Wilson Fellowship (declined), 2017.

UC Berkeley EECS Excellence Award (declined), 2017.

Tau Beta Pi, 2016.

Stanford University President's Award for Excellence in the Freshman Year, 2013.

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